Sangeet Paul

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Computational physicist and data scientist with 6+ years of research experience in gravitational-wave astronomy and Bayesian inference. Proven track record building high-performance scientific software, including JAX-accelerated population inference pipelines and GPU-optimized statistical tools, for LIGO collaboration research. Deep expertise in probabilistic modeling, hierarchical Bayesian methods, machine learning, and HPC, with a strong foundation in general relativity. Seeking roles at the intersection of physics, data science, and applied ML.

Education

University of Oregon — Eugene, OR
PhD, Astrophysics — *2019 — 2025*

- Advisor: Ben Farr

NISER, HBNI — Bhubaneswar, India
BSc-MSc, High-Energy Physics — *2013 — 2018*

Skills

- **Statistical Methods**: Bayesian statistics, MCMC, Hierarchical modeling, Normalizing flows, Digital signal processing, Cluster analysis
- **Machine Learning**: TensorFlow, Keras, scikit-learn, NumPyro
- **GPU & HPC**: JAX, CuPy, CUDA
- **Scientific Python Stack**: NumPy, SciPy, pandas, matplotlib, emcee
- **Domain-specific Libraries**: AstroPy, Bilby, PyCBC, GWPy
- **Programming Languages**: *fluent*: Python || *reading proficiency*: C, C++, R, Java
- **Dev Tools & Software**: VS Code, PyCharm, Jupyter, Git, LaTeX, Mathematica, MATLAB
- **Physics Domain Knowledge**: Gravitational-wave astronomy, General relativity, Quantum mechanics, Quantum field theory

Work Experience

Research Assistant — Jul 2021 — Jul 2025
Institute of Fundamental Sciences, University of Oregon — *Eugene, OR*

- Member, **LIGO Scientific Collaboration** (Feb 2020 — present)
- **Hierarchical mergers of binary black holes in dynamical astrophysical environments**: Built JAX-accelerated Bayesian population inference software to find evidence of hierarchical mergers of binary black holes in gravitational wave catalogs using the coagulation model for black hole mergers in dynamical astrophysical environments, such as globular clusters and active galactic nuclei' accretion disks.
- **Joint inference of astrophysical and noise parameters in gravitational wave data**: Built high-dimensional Bayesian inference software for time-series and frequency-series data from gravitational-wave interferometers to estimate astrophysical signal models and spline-based detector noise models simultaneously. Found localized correlations between signal waveform and noise power spectral density.
- **Clustering algorithms for significantly uncertain data**: Built algorithms to cluster data with high uncertainty, such as data from gravitational-wave interferometers. Found probabilistic clusters and outliers in mass–spin space.

Teaching Assistant — Sep 2019 — Jul 2025
Department of Physics, University of Oregon — *Eugene, OR*

- **Grad-level**: Scientific computation (machine learning), Design of experiments (Bayesian statistics).
- **Undergrad-level**: Quantum mechanics, Astrophysics, Astronomy, Fourier analysis, Introductory physics.

Research Fellow — 2018 — 2019
NISER, HBNI — *Bhubaneswar, India*

- **Blackfolds in higher-dimensional gravity**: Analyzed the stability of higher-dimensional equivalents of black holes. Verified through simulations that a perturbed black string asymptotically settles down to a black hole due to the Gregory–Laflamme instability.

Research Scholar 2013 — 2018
NISER, HBNI *Bhubaneswar, India*

- **Membrane – gravity duality in a large number of dimensions**: Demonstrated that black hole event horizons are analogous to hydrodynamic membranes.
- **Analytical predictions of the mass function of halos**: Analyzed the effects of varying dark matter halos' mass functions on astronomical and cosmological observations.
- **Magneto-optic Kerr effect: experiment design**: Constructed theory for an experiment to use the magneto-optic Kerr effect to analyze magnetized surfaces.

Research Scholar May — Jul 2017
Institute of Physics, HBNI *Bhubaneswar, India*

- **Superstring theory: Tree-approximation scattering amplitudes**: Studied the foundations of superstring theory, starting with bosonic strings, moving on to supersymmetry and gauge interactions, finishing with the evaluation of tree-level scattering amplitudes.

Research Scholar May — Jul 2016
Centre for Excellence in Basic Sciences, University of Mumbai *Mumbai, India*

- **Quantum field theory: Decays of the Higgs boson**: Studied relativistic quantum mechanics, quantum electrodynamics, Feynman diagrams, renormalization, and gauge field theories. Analyzed the radiation of gluon jets, the Coleman-Weinberg potential, and the decays of the Higgs boson.

Personal Projects

Simulations of semi-quantum cellular automata github.com/sangeetpaul/sqca

Solving chess problems to generate new integer sequences github.com/sangeetpaul/chess

Efficient integer-based encoding for binary trees github.com/sangeetpaul/ancestree

Documenting the endangered Gahaḷā language sangeetpaul.notion.site/gahala

Service Experience

Graduate Support Group, Dept of Physics, U Oregon 2023 — 2024

PhD Admissions Committee, Dept of Physics, U Oregon 2022 — 2023

Vice-President, Graduate Student Council, Dept of Physics, U Oregon 2021 — 2022

Publications

Google Scholar || **ORCID** || **NASA ADS**

Languages

English, Hindi, Odia, Bengali, Gahaḷā